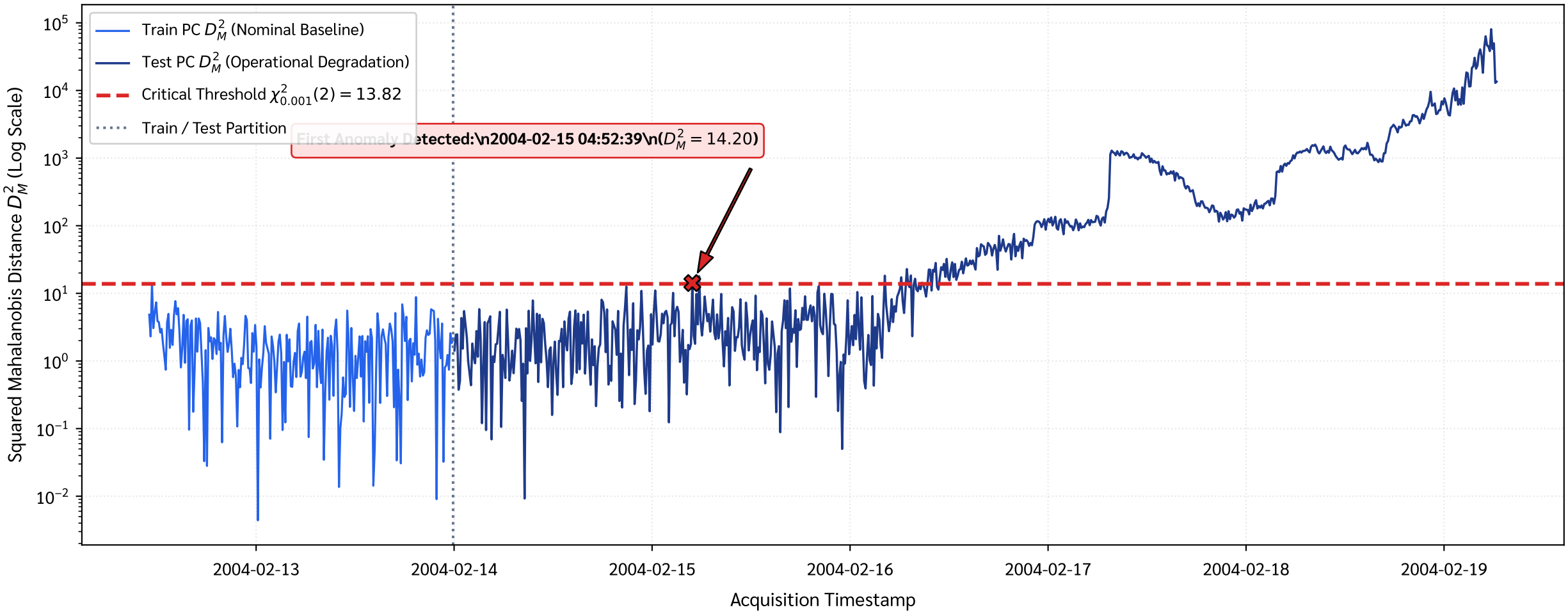
Figure 6: Latent-Variable Mahalanobis Distance ($T^2$) Anomaly Detection Timeline
Train PC $D_M^2$ (Nominal Baseline)
Test PC $D_M^2$ (Operational Degradation)
Critical Threshold $\chi^2_{0.001}(2) = 13.82$
Train / Test Partition
etected:\n2004-02-15 04:52:39\n($D_M^2 = 14.20$)
Squared Mahalanobis Distance $D_M^2$ (Log Scale)
$10^5$
$10^4$
$10^3$
$10^2$
$10^1$
$10^0$
$10^{-1}$
$10^{-2}$
2004-02-13
2004-02-14
2004-02-15
2004-02-16
2004-02-17
2004-02-18
2004-02-19
Acquisition Timestamp